University of Dhaka

Affiliated Faridpur Engineering College

Department of Computer Science and Engineering

1^{st} year semester 1^{st}, 2^{nd} in course Examination -2024 (Session: 2023-24)

Time: 50 mints **Marks: 20**

1. (a) If $y = x^n$ then show that $y_n = 2^n\{1.3.5.....(2n-1)\}x^n$;Where $n \in \aleph$ 3

(b) If $y = \sin nx + \cos nx$ then show that $y_r = n^r[1+(-1)^r \sin 2nx]^{1/2}$ 3

(c) What is called constant of integration? Evaluate, $(i)\int \frac{dx}{a\cos x + b\sin x}$ $(ii)\int \frac{\sin x dx}{\sqrt{1+\cos x}}$ $1+1.5\times 2$

2. (a) Find the interval where the function is concave up and concave down for $f(x) = 2x^3 - 9x^2 + 12x - 3$ 2

(b) Find the maximum and minimum values of u where $u = \frac{4}{x} + \frac{36}{y}$ and $x + y = 2$ 5

(c) A firm sells all product it makes at tk. 12.00 Per unit. The cost of making x units 3
$c(x) = 20 + 0.6x + 0.01x^2$. Find the maximum profit.